MIL-B-16129B(OS)
10 December 1974
SUPERSEDING
MIL-B-16129A
8 May 1953

MILITARY SPECIFICATION

BEARING, BALL, ANNULAR, OPEN TYPE (FOR TORPEDO GYROSCOPES)

This specification is approved for use by all the departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a high precision, open (or magneto), alloy steel ball bearing for use in torpedo gyroscopes.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents listed herein, the requirements of this specification shall apply.

SPECIFICATIONS

Federal

FF-B-171	Bearings, Ball, Annular (General Purpose)

Military

MIL-B-197	Bearing, Antifriction, Associated Part and Subassemblies, Packaging of
MIL-L-6085	Lubricating Oil, Instrument, Aircraft, Low Volatility

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STANDARDS

Federal

*FED-STD-66	Steel: Chemical Composition and Hardenability
*FED-STD-151	Metals: Test Methods

Military

*MIL-STD-129	Marking for Shipment and Storage
*MIL-STD-690	Failure Rate Sampling Plans and Procedures

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 First article. The ball bearing furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3 and 6.2).

3.2 Type. Except as modified herein, bearings shall comply with the requirements of FF-B-171, type B, grade I, (angular contact, single row, single mounting). Bearings shall be separable.

3.3 Assembly.

3.3.1 Dimensions. The bearings shall be of the dimensions shown in table I.

Table I

DIMENSIONS

Section	Dimension (inch)	Tolerance (inch)	
		Plus	Minus
Bore	0.2362	0.0000	0.0003
Outside diameter	.9449	.0004	.0000
Width (inner and outer race)	.2756	.000	.005

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3.3.2 Construction. The inner race, retainer, and balls shall be a matched assembly. The inner race, retainer, and balls need not be interchangeable. The outer race shall be interchangeable in all bearing assemblies. At no load, the contact angle of the balls with the outer race in a completely assembled bearing shall be 15° ± 4°. A bearing assembly shall contain eight 3/16-inch-diameter balls selected and graded so that the difference in diameter between the largest ball and the smallest ball in any assembled set shall not exceed 0.000025 inch. The retainer and balls shall be separable from the inner and outer race; and, when so removed, the balls shall be securely held by the retainer. Balls shall be capable of removal from the retainer without damage to the balls or the retainer.

3.3.3 Eccentricity.

3.3.3.1 Inner race. The eccentricity of the inner race at its smallest diameter and at circles not less than 0.012 inch either side of the smallest diameter shall not exceed 0.0002 inch total indicator reading when tested as specified in 4.5.2.1.

3.3.3.2 Outer race. The eccentricity of the outer race shall not exceed 0.0004 inch total indicator reading when tested as specified in 4.5.2.2.

3.3.4 Face runout. The face (axial) runout of the outer race shall not exceed 0.0006 inch total indicator reading when tested as specified in 4.5.3.

3.3.5 End shake. When tested as specified in 4.3.4, the end faces of the inner race shall lie in planes not more than 0.005 inch distant from the planes established by the adjacent end faces of the outer race.

3.3.6 Sensitivity. A moment of 0.66 gramcentimeter shall start rotation, in either direction, when tested as specified in 4.5.5.

3.4 Components.

3.4.1 Surface imperfections. The running surfaces of the races and the balls shall have a highly polished, bright finish and shall be free from tool marks, chatter waves, grinding scratches, pits, rust, and other surface imperfections visible with a 2-power magnifier.

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3.4.2 Balls. The balls shall be made of alloy steel hardened to Rockwell "C" 61 to 65 inclusive. The nominal ball size shall be 3/16 inch and the maximum variation in sphericity shall be 0.000025 inch. The exact ball size shall be controlled by the manufacturer to produce bearings which shall comply with the requirements of this specification.

3.4.3 Races. The races shall be made of alloy steel hardened to Rockwell "C" 61 to 65 inclusive.

3.4.4 Retainers. Unless otherwise specified, retainers shall be made from steel. Retainers shall be smoothly finished, free from fins, chips, or wire edges. The ball indentations in the side walls of the retainer shall be concentric for any pair with respect to the center line of the ball pocket opening. Retainers shall clear the races at all times.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspections. The inspection requirements herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

*4.3 First article inspection. Unless otherwise specified (see 6.2), first article samples shall be selected in accordance with 4.3.1 and subjected to all the examinations and tests specified herein. Failure of the samples to meet all the requirements shall be cause for rejection of the first article sample.

*4.3.1 Sampling for first article inspection. Twenty bearings shall be randomly selected from the first quantity of bearings made after contract award.

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*4.4 Quality conformance inspection. Quality conformance inspection for acceptance of individual lots shall consist of examinations and tests for all the requirements specified in sections 3 and 5.

*4.4.1 Sampling and testing for quality conformance inspection. Sampling and testing shall be in accordance with MIL-STD-690, FED-STD-151, and as specified herein.

4.5 Test methods.

*4.5.1 Hardness of balls and races. Tests shall be conducted in accordance with FED-STD-66.

4.5.2 Eccentricity.

4.5.2.1 Inner race. The eccentricity shall be measured at right angles to the bearing surface at each of the three diameters specified in 3.3.3.1, as the inner race is rotated about centers true with the axis of the bore.

4.5.2.2 Outer race. The eccentricity of the outer race shall be measured with the assembled bearing under light axial thrust load. The outer race shall be rotated, the inner race being fixed upon a stationary arbor.

4.5.3 Face runout. The bearing shall be assembled and tested under a light axial thrust load. The face (axial) runout of the outer race shall be measured with the outer race held stationary (not rotating) as the inner race is rotated upon centers true with the axis of the bore.

4.5.4 End shake. The assembled bearing shall be subjected to an axial thrust load of 2 pounds and tested for the requirements specified in 3.3.5.

4.5.5 Sensitivity. Bearing parts shall be fully seated and then backed off 0.004 inch. The bearing shall be subjected to a radial load of 1-1/2 pounds and a moment of 0.66 gramcentimeter shall be applied in both directions of rotation.

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4.6 Rejection and resubmission. The lot shall be subject to rejection if any specimen fails to pass any of the specified tests. A lot that has been rejected may be resubmitted for Government inspection provided all deficiencies have been corrected.

5. PREPARATION FOR DELIVERY

5.1 General. Unless otherwise specified, bearings shall be prepared for delivery in accordance with MIL-B-197, except as follows.

*5.2 Preservation and packaging.

*5.2.1 Level A.

5.2.1.1 Demagnetizing. All bearings shall be demagnetized prior to cleaning.

*5.2.1.2 Cleaning. The ball bearings shall be cleaned in accordance with the method specified in MIL-B-197.

5.2.1.3 Preservation. Unless otherwise specified, lubricating oil complying with MIL-L-6085 shall be used.

5.2.1.4 Packaging. The number of bearings specified by the contract shall be immersed in the preservative, in an oil-resistant, transparent vial, allowing 5 percent outage. Ten vials shall be placed in a snug fitting carton.

*5.2.2 Level C. When this level is required each item shall be cleaned, preserved, and packaged in accordance with the manufacturer's commercial practice.

*5.3 Packing.

*5.3.1 Level A. The ball bearings preserved and packaged as specified in 5.2.1.3 and 5.2.1.4 shall be packed as specified in MIL-B-197.

*5.3.2 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the

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point of delivery. Containers shall conform to the rules and regulations of the common carrier, as applicable to the mode of transportation. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.4 Marking of package. The date of preservation shall be shown on each individual package.

*5.5 Marking for shipment. In addition to any special marking required by the contract or order, packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

*6.1 Intended use. The alloy steel ball bearings covered by this specification are intended for use in torpedo gyroscopes.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Quantity desired

c. Whether first article is required (see 3.1, 4.3 and 6.2.1)

d. Number of bearings to be placed in a vial (see 5.2.1.4)

e. Levels of preservation, packaging, and packing required (see 5.2 and 5.3).

*6.2.1 Whether first article is required. When a supplier has been in continuous production from contract to contract, consideration should be given to waive the first article inspection. If required, indicate:

a. Where the first article inspection is to be conducted (at the supplier's plant or at a Government plant).

b. That approval of the first article samples or the waiving of first article inspection shall not relieve the supplier of his obligation to fulfill all other requirements of this specification and contract.

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6.3 Bearing units. The bearing shall be treated as a unit or matched assembly for service use.

*6.4 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE: MIL-B-16129B(OS), Bearings, Ball, Annular, Open Type (For Torpedo Gyroscopes)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
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